INCH-POUND

A-A-55511
June 10, 1994

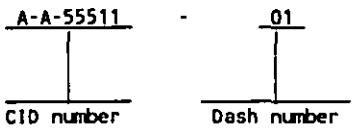
COMMERCIAL ITEM DESCRIPTION

MULTIMETER, 3 1/2 DIGIT, DIGITAL DISPLAY, HAND HELD

The General Services Administration has authorized the use of this commercial item description (CID).

Abstract. This CID covers the general requirements for a 3 1/2 digit, digital display, hand held multimeter. Multimeters covered by this CID are intended for commercial/industrial applications.

Part or Identifying Number (PIN). The PIN for the CID shall be as shown in the following example:




Salient characteristics.

Readout display. The readout display shall include a Liquid Crystal Display (LCD) having a digital readout of 3,200 counts resolution, updated at least 2.5 times per second. A 31 segment analog bar graph readout shall be included in the display, updating at least 25 times per second. Annunciators shall clearly indicate low battery condition, function, and range.

Automatic polarity. The polarity of the measured value shall be automatically indicated on the digital and analog displays.

Range and function control. This multimeter shall be supplied with a single rotary switch to select all functions (voltage, current, resistance, and diode/continuity test). Ranging shall be automatic (with a manual override) in dc/ac voltage, dc/ac current, and resistance.

Maximum voltage to earth. The maximum voltage to earth shall be 1000 volts dc, 750 volts ac rms (sine).

Electrical power source. This multimeter shall operate from an internal battery capable of sustaining 2,000 hours of operation (e.g., alkaline). The unit may be shipped with supplied battery installed.

Battery type. This multimeter shall operate from battery power.

Sleep mode. An automatic shut-off (sleep mode) shall occur after 1 hour of continuous switch inactivity (20 minutes in diode test).

Multimeter dimensions. This multimeter shall have a maximum height of 28 millimeters (mm) (1.12 inches), a maximum width of 75 mm (2.95 inches), and a maximum length of 166 mm (6.55 inches).

Weight. The weight of this multimeter shall be 340 grams (12 ounces) maximum.

Safety. This multimeter shall be designed and listed to UL 1244.

Design and construction. Multimeters supplied to this CID shall be as specified herein and shall meet the requirements of MIL-T-28800 for a style B, class 2 instrument, and UL 1244.

AMSC N/A

FSC 6625

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:50Z
Check the source to verify that this is the current version before use.

AC voltage ranges, resolution, and accuracy. The multimeter shall provide the resolution and accuracy for ac voltage measurements as specified in table II.

TABLE II. AC voltage resolution and accuracy.

| AC voltage range | Resolution | Accuracy |
|---|---|---|
| 45-500 Hz | | ±(2% + 2 digits) |
| 3.2 V | 10 mV | |
| 45-1 kHz | | |
| 32.00 V | 10 mV | |
| 320.0 V | 100 mV | |
| 750 V | 1 V | |

Input impedance (ac voltage). The input impedance shall be greater than or equal to 10 megohms (MΩ), paralleled by less than 50 picofarads, on all ranges.

Overload protection (ac voltage). The multimeter shall be capable of continuously withstanding, without damage or blown fuse, 750 V ac rms or 1000 V dc on all ranges.

AC voltage common mode rejection ratio. The ac voltage common mode rejection ratio with a 1000Ω unbalance in the low lead, shall be not less than 60 decibels, dc to 60 Hz.

DC current ranges, resolution, accuracy, and burden voltage. The multimeter shall provide continuous dc current measurement accuracy and resolution within and including the rated burden voltage as specified in table III.

TABLE III. DC current ranges, resolution, accuracy and burden voltage.

| DC current range | Resolution | Accuracy | Burden voltage |
|---|---|---|---|
| 32.00 mA | 0.01 mA | ±(1.5% +2 digit) | 6 mV/mA |
| 320.0 mA | 0.1 mA | | |
| 10 A | 10 mA | | 50 mV/A |

Overload protection (dc current). The multimeter shall be capable of continuously withstanding, without damage, 600 mA on all ranges, except that the 10 A range shall be capable of withstanding 10 A continuously, and 20 A for 30 seconds. All current inputs shall be fused or protected to interrupt overload currents and overloads in circuits.

AC current ranges, resolution, accuracy, and burden voltage. The multimeter shall provide continuous ac current measurement accuracy and resolution within and including the rated burden voltage as specified in table IV.

TABLE IV. AC current ranges, resolution, accuracy and burden voltage.

| AC current range | Resolution | Accuracy | Burden voltage |
|---|---|---|---|
| 32 mA | 0.01 mA | ±(2.5% +2 digit) | 6 mV/mA |
| 320 mA | 0.1 mA | | |
| 10 A | 10 mA | | 50 mV/A |



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:50Z
Check the source to verify that this is the current version before use.

Touch hold. This multimeter shall have the switch selectable capacity of acquiring and holding, without operator intervention, a measurement whose reading remains in the display for viewing after the test signal disappears or after the test leads have been removed from the device under test. A new measurement shall automatically be acquired and held when an input is present and becomes stable. An audible beep shall indicate the new measurement.

Technical documentation. Multimeters shall be provided with technical documentation including owner's manual, and related information on operation and use.

Marking. Multimeters supplied to this CID shall be marked with the manufacturer's standard commercial PIN and the UL 1244 listing mark.

Regulatory requirements. This section is not applicable to this CID.

Quality assurance provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection, examination, and test requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections, examinations, or tests set forth in this description where such inspections, examinations, and tests are deemed necessary to assure supplies and services conform to prescribed requirements.

Contractor certification statement. The contractor shall certify and maintain objective quality evidence that the product offered meets the requirements of this CID, and that the product conforms to the producer's own drawings, specifications, standards, quality assurances practices, and is the same as the product provided as a bid sample. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Certificate of compliance. A certificate of compliance shall accompany all digital display multimeters supplied to this CID. The letter of certification shall state that the digital display multimeters meets all the requirements of UL 1244, and all requirements of this CID.

Packaging.

Preservation, packaging, packing, labeling, and marking. Preservation, packaging, packing, labeling, and marking shall be as specified in the contract or purchase order.

Notes. This section contains relevant information which is useful to buyers, users, and suppliers in the process of acquiring the item, but is not mandatory.

Government users. To acquire information on obtaining this multimeter from the Government inventory system, please contact Defense Electronics Supply Center, ATTN: DESC-EOE, 1507 Wilmington Pike, Dayton, OH 45444-5521, or telephone (513) 296-5168 and ask for A-A-55511-01.

Referenced documents. The following documents, of the issue in effect on date of invitation for bids for request for proposal, form a part of this CID to the extent specified herein.

Military Specification

MIL-T-28800 - Test Equipment For Use With Electrical And Electronic Equipment, General Specification For.

(Copies of the military specification are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

UNDERWRITERS LABORATORIES, INCORPORATED (UL)

UL 1244 - Electrical and Electronic Measuring and Testing Equipment, Safety Standard For.

(Applications for copies should be addressed to the Underwriters Laboratories, Incorporated, 333 Pfingsten Road, Northbrook, IL 60062-2096.)



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:50Z
Check the source to verify that this is the current version before use.

Fusing. The milliampere range shall be fused at 630 milliamperes (mA) and 250 volts (V), with an interrupt rating of 1500 amperes (A). The 10-ampere current measuring circuitry shall be fused at 15 A, 600 V, and shall be capable of interrupting peak currents of 10,000 A without damage to the multimeter or hazard to the operator.

Test leads. The multimeter shall be supplied with a set of polyvinyl chloride (PVC) insulated test leads with safety shrouded right angle inputs. These test leads shall be 1.2 meters (48 inches) minimum and rated at 1500 V, 10 A maximum (one red and one black in color).

Operating temperature. The multimeter shall perform to its full specified accuracy from +18°C to +28°C, and all ranges and functions from +0°C to +50°C.

Storage temperature. The storage temperature of this multimeter shall be -40°C to +60°C.

Temperature coefficient. Below +18°C and above +28°C, the accuracy specifications shall be degraded by no worse than 0.1 times the specified accuracy per °C.

Relative humidity. The relative humidity shall be 0 percent to 90 percent for the temperature range of +0°C to +35°C, 0 percent to 70 percent for the temperature range of +35°C to +50°C.

Shock and vibration. The shock and vibration requirements shall be in accordance with MIL-T-28800 for a style B, class 2 instrument.

DC voltage ranges, resolution, and accuracy. The multimeter shall provide the accuracy and resolution for dc voltage measurements as specified in table I.

DC/ac voltage, dc/ac, current, and ohms accuracy. The accuracy which is indicated in the dc/ac voltage, dc/ac current tables, and the ohm table is specified as ± ([percentage of reading] + [number of least significant digits]) at +18°C to +28°C (+64°F to +82°F with relative humidity to 90 percent for a period of 1 year after calibration.

TABLE I. DC voltage resolution and accuracy.

| DC voltage range | Resolution | Accuracy |
|---|---|---|
| 320 millivolts (mV) | 0.1 mV | ±(0.4% + 1 digit) |
| 3.200 V | 1 mV | |
| 32.00 V | 10 mV | |
| 320.0 V | 100 mV | |
| 1000 V | 1 V | |

Input impedance (dc voltage). The input impedance shall be equal to or greater than 10 MΩ on any range.

Overload protection (dc voltage). The multimeter shall be capable of continuously withstanding, without damage or blown fuse, 1000 V dc or 750 V ac rms on all ranges except the mV range where the protection shall be 500 V dc or 500 V ac rms.

DC voltage normal mode rejection ratio. The dc voltage normal mode rejection ratio shall be not less than 60 decibels at dc 50 hertz (Hz) or 60 Hz.

DC voltage common mode rejection ratio. The dc common mode rejection ratio, with a 1000 ohms (Ω) unbalance in the low lead, shall not be less than 120 decibels at dc, 50 Hz or 60 Hz.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:50Z
Check the source to verify that this is the current version before use.

Overload protection (ac current). The multimeter shall be capable of continuously withstanding, without damage, 600 mA on all ranges; except the 10 A range shall be capable of withstanding 10 A continuously. All current inputs shall be fused or protected to interrupt overload currents and overloads in circuits up to 600 V.

Ohms ranges, resolution, and accuracy. The multimeter shall provide the resolution, and accuracy for resistance measurements as specified in table V.

TABLE V. Ohms and diode test accuracy.

| Ohms range | Resolution | Accuracy |
|---|---|---|
| 320.0Ω | 0.1Ω | ±(0.5% + 2 digits) |
| 3.2 kΩ | 1Ω | ±(0.5% + 1 digit) |
| 32.00 kΩ | 10Ω | |
| 320.0 kΩ | 100Ω | |
| 3.200 MΩ | 1 kΩ | |
| 32.00 MΩ | 10 kΩ | ±(2% + 1 digit) |

Overload protection (ohms). The multimeter shall be capable of continuously withstanding, without damage or blown fuse, 500 V dc or rms on all ranges.

Maximum current through unknown resistance. The multimeter shall not provide more than 0.75 mA on any range.

Maximum open circuit voltage. The maximum open circuit voltage shall not be greater than 3.1 V on any range.

Continuity test. The multimeter shall indicate a nominal conducting path of 150Ω or less , of 70 milliseconds or longer duration, by means of a continuous audible tone.

Diode test. The multimeter shall be capable of displaying diode or transistor junction voltage drops to at least 2.0 V. A momentary audible tone shall indicate a forward biased junction, and the absence of sound shall indicate a reversed biased or open junction.

Maximum test current. The maximum diode or continuity test current shall be 900 microamperes.

Overload protection (diode or continuity test). The multimeter shall be capable of continuously withstanding, without damage or blown fuse, 500 V dc rms when in diode test or continuity test function.

Calibration and maintenance. The design of the multimeter shall provide for readily accessible calibration, accomplished by adjustment of a single variable-value component. The calibration by substitution of selected components or parts is unacceptable. The calibration interval shall be a period of 1 year or greater, based on an operating time of 2000 hours. Other than the precision calibration source, maintenance and calibration shall require only ordinary general-purpose test tools.

Response time. Response time is defined as the time interval between an abrupt change of input and the time the new value is displayed within specified accuracy, including the worst case time to change ranges in the autorange mode.

DC response. Direct current or voltage response shall be less than or equal to 1 second.

AC response. Alternating current or voltage response time shall be less than or equal to 2 seconds.

Ohms response. Resistance response time shall be less than or equal to 1 second on all ranges, except that in the highest resistance ranges it shall be not more than 2 seconds in the 3.2 MΩ range and 10 seconds in the 32 MΩ range.



Source: https://assist.dla.mil -- Downloaded: 202640-26T09:50Z
Check the source to verify that this is the current version before use.

Ordering data. The contract or purchase order shall specify the following:

a. CID document number and revision and CID PIN.

b. Quality assurance provisions.

c. Packaging requirements.

Comments. Comments on this CID should be directed to Defense Electronics Supply Center, ATTN: DESC-ELDM, 1507 Wilmington Pike, Dayton, OH 45444-5765, or telephone (513) 296-6189, or facsimile (FAX) (513) 296-8868.

Suggested source of supply. A suggested source of supply is listed herein. Additional sources will be added as they become available.

| A-A-55511- | Vendor commercial PIN CAGE 89536 |
|---|---|
| 01 | 75-2 |

| Vendor CAGE number | Vendor name and address |
|---|---|
| 89536 | Fluke Corporation<br>P.O. Box 9090<br>Everett, WA 98206-9090<br>(206) 356-5443 |

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - 7FXE

PREPARING ACTIVITY:

DLA-ES

Project 6625-D056



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:50Z
Check the source to verify that this is the current version before use.